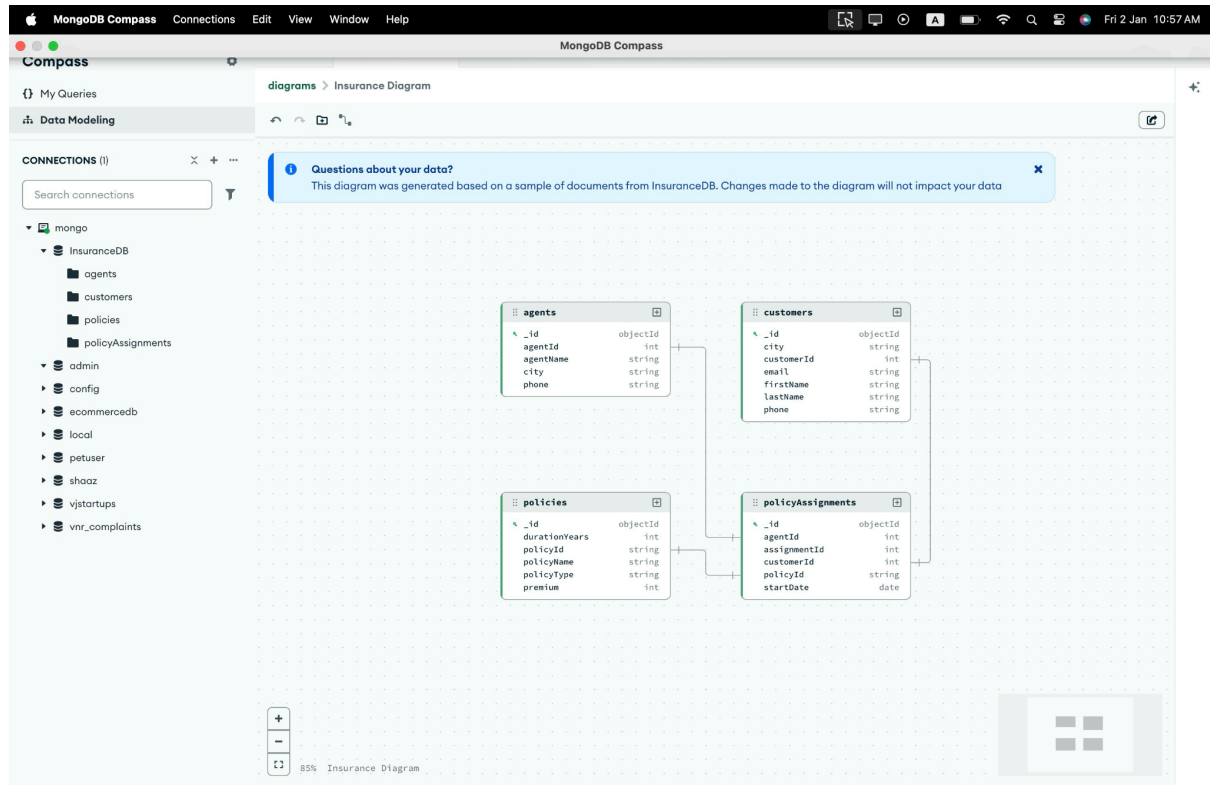


Day - 5 MongoDB Assignment

By shaaz



1.Creating DB:-

use InsuranceDB

2.Inserting and creating collections:-

i:-Customers collection:-

```
db.customers.insertMany([
{
  customerId: 1,
  firstName: "Amit",
  lastName: "Sharma",
  email: "amit@gmail.com",
  phone: "9876543210",
```

```
    city: "Hyderabad"
  },
  {
    customerId: 2,
    firstName: "Sara",
    lastName: "Khan",
    email: "sara@gmail.com",
    phone: "9876501234",
    city: "Mumbai"
  },
  {
    customerId: 3,
    firstName: "John",
    lastName: "Doe",
    email: "john@gmail.com",
    phone: "9123456789",
    city: "Bangalore"
  }
])
```

ii:-Inserting Agents and creating Agents collections:-

```
db.agents.insertMany([
  {
    agentId: 101,
    agentName: "Ravi Kumar",
    city: "Hyderabad",
    phone: "9000011111"
  },
  {
    agentId: 102,
    agentName: "Neha Singh",
    city: "Delhi",
    phone: "9000022222"
  }
])
```

iii:-Inserting and creating the policies:-

```
db.policies.insertMany([
  {
    policyId: "P1001",
    policyName: "Health Plus",
    policyType: "Health",
```

```
    premium: 12000,  
    durationYears: 5  
  },  
  {  
    policyId: "P1002",  
    policyName: "Life Secure",  
    policyType: "Life",  
    premium: 15000,  
    durationYears: 10  
  }  
])
```

iv:-Policy Assignment collection and insertion into the collection:-

```
db.policyAssignments.insertMany([  
  {  
    assignmentId: 1,  
    customerId: 1,  
    policyId: "P1001",  
    agentId: 101,  
    startDate: new Date("2024-01-01")  
  },  
  {  
    assignmentId: 2,  
    customerId: 2,  
    policyId: "P1002",  
    agentId: 102,  
    startDate: new Date("2023-06-15")  
  }  
])
```

```
Terminal Shell Edit View Window Help
shaazhussain - mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 - ?directConnection=true&__CFBundleIdentifier=com.apple.Terminal - 208x65

insuranceddb> show collections;

insuranceddb> use InsuranceDB
switched to db InsuranceDB
InsuranceDB> show collections;
agents
customers
policies
policyAssignments
InsuranceDB> db.customers.find().pretty();
[
  {
    _id: ObjectId('69574ec459a93b3960ae3c63'),
    customerId: 1,
    firstName: 'Amit',
    lastName: 'Sharma',
    email: 'amit@gmail.com',
    phone: '9876543210',
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('69574ec459a93b3960ae3c64'),
    customerId: 2,
    firstName: 'Sara',
    lastName: 'Khan',
    email: 'sara@gmail.com',
    phone: '9876501234',
    city: 'Mumbai'
  },
  {
    _id: ObjectId('69574ec459a93b3960ae3c65'),
    customerId: 3,
    firstName: 'John',
    ...
  }
]

InsuranceDB> db.policyAssignments.aggregate([
... {
...   $lookup: {
...     from: 'customers',
...     localField: 'customerId',
...     foreignField: 'customerId',
...     as: 'customerDetails'
...   }
... })
... ]
[
  {
    _id: ObjectId('69574ed259a93b3960ae3c6a'),
    assignmentId: 1,
    customerId: 1,
    policyId: 'P1001',
    agentId: 01,
    startDate: ISODate('2024-01-01T00:00:00.000Z'),
    customerDetails: [
      {
        _id: ObjectId('69574ec459a93b3960ae3c63'),
        customerId: 1,
        firstName: 'Amit',
        lastName: 'Sharma',
        email: 'amit@gmail.com',
        phone: '9876543210',
        city: 'Hyderabad'
      }
    ]
  }
]
```

3.For displaying the collections:-

show collections

```
Terminal Shell Edit View Window Help
shaazhussain - mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 - ?directConnection=true&__CFBundleIdentifier=com.apple.Terminal - 208x65

> 1 | dhow collections;
> 2 |
^

insuranceddb> show collections;

insuranceddb> use InsuranceDB
switched to db InsuranceDB
InsuranceDB> show collections;
agents
customers
policies
policyAssignments
InsuranceDB> db.customers.find().pretty();
[
  {
    _id: ObjectId('69574ec459a93b3960ae3c63'),
    customerId: 1,
    firstName: 'Amit',
    lastName: 'Sharma',
    email: 'amit@gmail.com',
    phone: '9876543210',
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('69574ec459a93b3960ae3c64'),
    customerId: 2,
    firstName: 'Sara',
    lastName: 'Khan',
    email: 'sara@gmail.com',
    phone: '9876501234',
    ...
  }
]
```

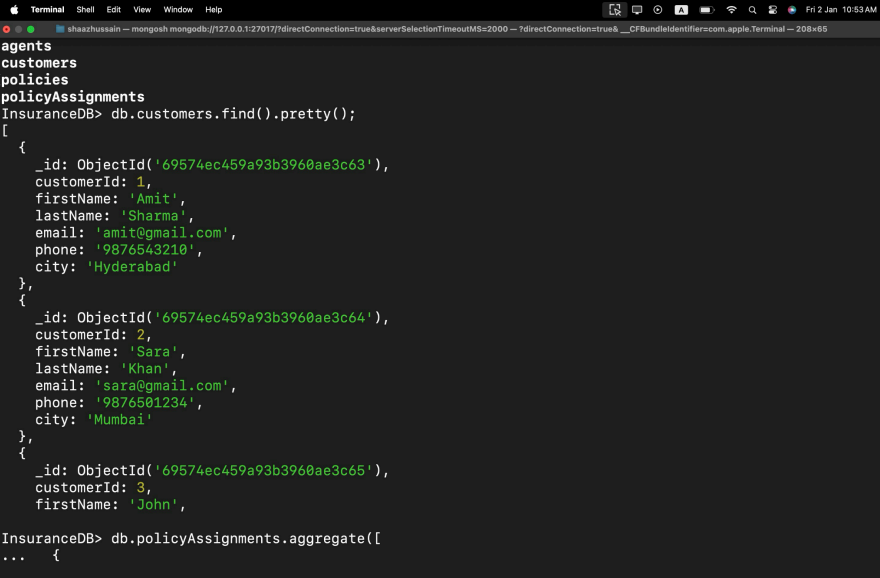
4. For showing the records in the Collections:-

db.customers.find()

db.agents.find()

db.policies.find()

db.policyAssignments.find()

A screenshot of a macOS Terminal window. The title bar shows 'Terminal' and standard window controls. The terminal text shows a list of collections: agents, customers, policies, and policyAssignments. The user enters 'InsuranceDB> db.customers.find().pretty();'. The output shows three customer records as JSON objects. The first record is for Amit Sharma, the second for Sara Khan, and the third for John. The terminal text is as follows:

```
agents
customers
policies
policyAssignments
InsuranceDB> db.customers.find().pretty();
[
  {
    _id: ObjectId('69574ec459a93b3960ae3c63'),
    customerId: 1,
    firstName: 'Amit',
    lastName: 'Sharma',
    email: 'amit@gmail.com',
    phone: '9876543210',
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('69574ec459a93b3960ae3c64'),
    customerId: 2,
    firstName: 'Sara',
    lastName: 'Khan',
    email: 'sara@gmail.com',
    phone: '9876501234',
    city: 'Mumbai'
  },
  {
    _id: ObjectId('69574ec459a93b3960ae3c65'),
    customerId: 3,
    firstName: 'John',
    ...
  }
]
```

4.Filtering Queries:-

```
i.db.customers.find({ city: "Hyderabad" });
```

```

Terminal Shell Edit View Window Help
shashussain — mongohq mongod: /j127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 — 7directConnection=true __CFBundleIdentifier=com.apple.Terminal — 208x65
shashussain@shashussain:~$ mongo --host 127.0.0.1 --port 27017 --directConnection=true
...
{
  _id: ObjectId('69574ec459a93b3960ae3c63'),
  customerId: 1,
  firstName: 'Amit',
  lastName: 'Sharma',
  email: 'amit@gmail.com',
  phone: '9876543210',
  city: 'Hyderabad'
}
InsuranceDB>

```

```
ii.db.policies.find({ premium: { $gt: 13000 } });
```

```

Terminal Shell Edit View Window Help
❏ @ shashishashank ~ % cd /Users/shashank/Downloads/ ; curl -s https://raw.githubusercontent.com/0x00sec/0x00sec/master/0x00sec.js | node
❏ {
  -id: ObjectId('69574acc459a93b396d8ae3c63'),
  customerId: 1,
  firstName: 'Ankit',
  lastName: 'Sharma',
  email: 'ankit@gmail.com',
  phone: '9876543218',
  city: 'Hyderabad'
}
❏ InsuranceDB.db.policies.find({ premium: { $gt: 15000 } });
❏ {
  -id: ObjectId('69574acc459a93b396d8ae3c6d'),
  policyId: 'P1802',
  policyName: 'Life Secure',
  policyType: 'Life',
  premium: 15000,
  durationYears: 10
}
❏ InsuranceDB

```

4.Sorting and Limit:-

i:-db.customers.find().sort({ firstName: 1 })

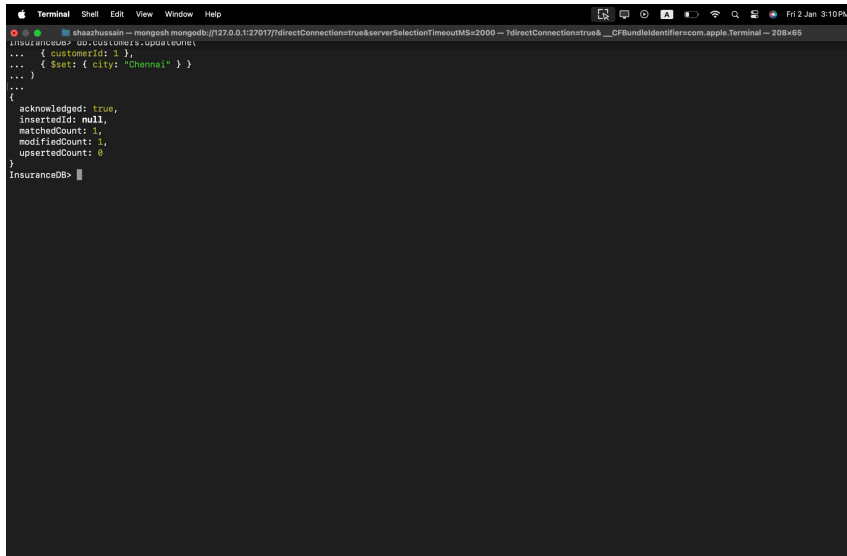
```
Terminal Shell Edit View Window Help
shaazhussain - mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 - Fri 2 Jan 2:19 PM
InsuranceDB> db.customers.find().sort({ firstName: 1 })
...
[
  {
    _id: ObjectId('69574ec459a93b3960ae3c63'),
    customerId: 1,
    firstName: 'Amit',
    lastName: 'Sharma',
    email: 'amit@gmail.com',
    phone: '9876543210',
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('69574ec459a93b3960ae3c65'),
    customerId: 3,
    firstName: 'John',
    lastName: 'Doe',
    email: 'john@gmail.com',
    phone: '9123456789',
    city: 'Bangalore'
  },
  {
    _id: ObjectId('69574ec459a93b3960ae3c64'),
    customerId: 2,
    firstName: 'Sara',
    lastName: 'Khan',
    email: 'sara@gmail.com',
    phone: '9876501234',
    city: 'Mumbai'
  }
]
InsuranceDB>
```

ii:-db.policies.find().sort({ premium: -1 }).limit(1):-

```
Terminal Shell Edit View Window Help
shaazhussain - mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 - Fri 2 Jan 2:20 PM
InsuranceDB> db.policies.find().sort({ premium: -1 }).limit(1)
...
[
  {
    _id: ObjectId('69574ecd59a93b3960ae3c69'),
    policyId: 'P1002',
    policyName: 'Life Secure',
    policyType: 'Life',
    premium: 15000,
    durationYears: 10
  }
]
InsuranceDB>
```

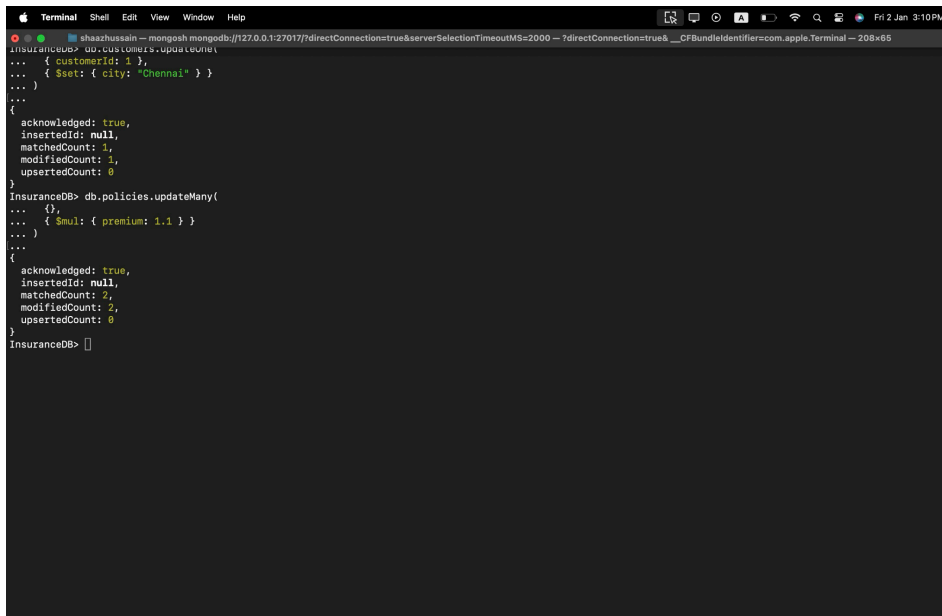
5.Update Queries:-

```
db.customers.updateOne(  
  { customerId: 1 },  
  { $set: { city: "Chennai" } }  
)
```



```
Terminal Shell Edit View Window Help
shaazhussain - mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 - ?directConnection=true&..._CFBundleIdentifier=com.apple.Terminal - 208x65
InsuranceDB> db.customers.updateOne(
... { customerId: 1 },
... { $set: { city: "Chennai" } }
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
InsuranceDB>
```

```
db.policies.updateMany(  
  {},  
  { $mul: { premium: 1.1 } }  
)
```



```
Terminal Shell Edit View Window Help
shaazhussain - mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 - ?directConnection=true&..._CFBundleIdentifier=com.apple.Terminal - 208x65
InsuranceDB> db.customers.updateOne(
... { customerId: 1 },
... { $set: { city: "Chennai" } }
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
InsuranceDB> db.policies.updateMany(
... {},
... { $mul: { premium: 1.1 } }
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
InsuranceDB>
```


5.Delete Query:-

db.customers.deleteOne({ customerId: 3 })

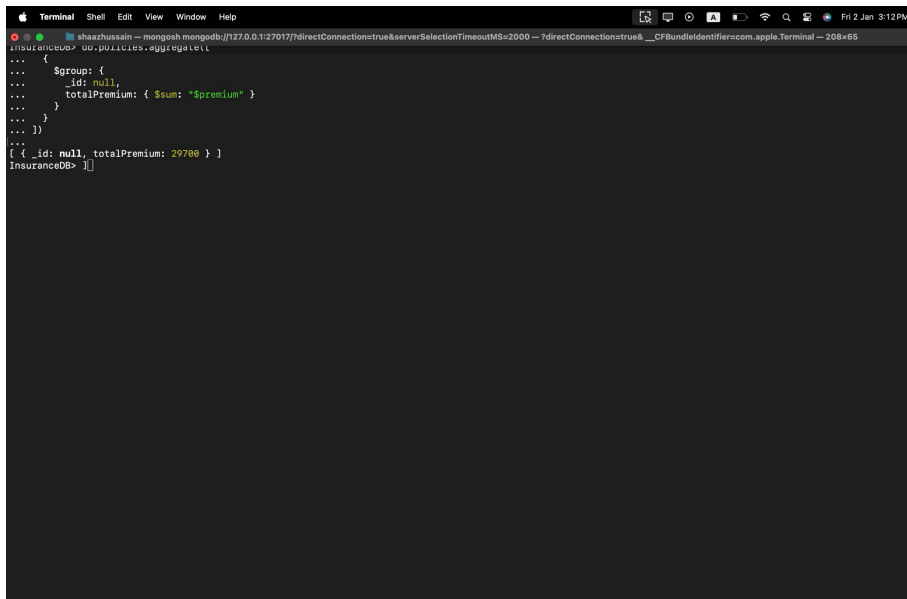
```
Terminal Shell Edit View Window Help
shahzhusain - mongosh mongod://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 - ?directConnection=true&_CFBundleIdentifier=com.apple.Terminal - 208x65
InsuranceDB> db.customers.updateOne(
... { customerId: 1 },
... { $set: { city: "Chennai" } }
... )
...
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
InsuranceDB> db.policies.updateMany(
... {},
... { $mul: { premium: 1.1 } }
... )
...
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
InsuranceDB> db.customers.deleteOne({ customerId: 3 })
...
{ acknowledged: true, deletedCount: 1 }
InsuranceDB>
```

6.Aggregation Queries:-

```
db.policies.aggregate([
{
  $group: {
    _id: "$policyType",
    avgPremium: { $avg: "$premium" }
  }
}]
)
```

```
Terminal Shell Edit View Window Help
shahzhusain - mongosh mongod://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000 - ?directConnection=true&_CFBundleIdentifier=com.apple.Terminal - 208x65
InsuranceDB> db.customers.updateOne(
... { customerId: 1 },
... { $set: { city: "Chennai" } }
... )
...
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
InsuranceDB> db.policies.updateMany(
... {},
... { $mul: { premium: 1.1 } }
... )
...
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
InsuranceDB> db.customers.deleteOne({ customerId: 3 })
...
{ acknowledged: true, deletedCount: 1 }
InsuranceDB> db.policies.aggregate([
... {
...   $group: {
...     _id: "$policyType",
...     avgPremium: { $avg: "$premium" }
...   }
... })
...
...
{ _id: 'Life', avgPremium: 16500 },
{ _id: 'Health', avgPremium: 13200.000000000002 }
]
InsuranceDB>
```

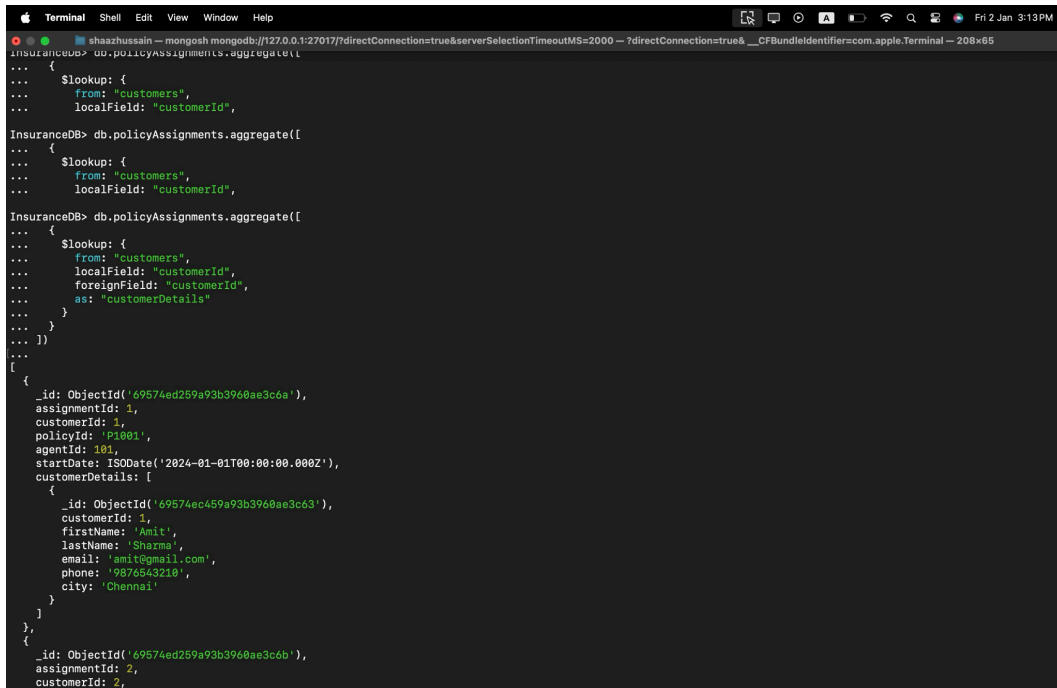
```
db.policies.aggregate([
  {
    $group: {
      _id: null,
      totalPremium: { $sum: "$premium" }
    }
  }
])
```

A screenshot of a macOS Terminal window. The title bar shows 'Terminal' and standard macOS window controls. The terminal text shows a MongoDB aggregation command being executed in the 'InsuranceDB' database. The output is a JSON array with one object containing a 'totalPremium' field with a value of 29780. The prompt 'InsuranceDB>' is visible at the bottom.

```
InsuranceDB> db.policies.aggregate([
... {
...   $group: {
...     _id: null,
...     totalPremium: { $sum: "$premium" }
...   }
... })
... ]
[ { _id: null, totalPremium: 29780 } ]
InsuranceDB>
```

7.Lookup :-

```
db.policyAssignments.aggregate([
{
  $lookup: {
    from: "customers",
    localField: "customerId",
    foreignField: "customerId",
    as: "customerDetails"
  }
}
])
```

A screenshot of a macOS Terminal window with a dark background. The window title is "Terminal" and the menu bar shows "Terminal", "Shell", "Edit", "View", "Window", and "Help". The terminal shows a MongoDB aggregation command being executed in the "InsuranceDB" database. The command is `db.policyAssignments.aggregate([{$lookup: {from: 'customers', localField: 'customerId', foreignField: 'customerId', as: 'customerDetails'}}])`. The output is a JSON array with two documents. The first document has `_id`: `ObjectId('69574ed259a93b3960ae3c6a')`, `assignmentId`: `1`, `customerId`: `1`, `policyId`: `'P1001'`, `agentId`: `101`, `startDate`: `ISODate('2024-01-01T00:00:00.000Z')`, and `customerDetails`: an array containing one object with `_id`: `ObjectId('69574ec459a93b3960ae3c63')`, `customerId`: `1`, `firstName`: `'Amit'`, `lastName`: `'Sharma'`, `email`: `'amit@gmail.com'`, `phone`: `'9876543210'`, and `city`: `'Chennai'`. The second document has `_id`: `ObjectId('69574ed259a93b3960ae3c6b')`, `assignmentId`: `2`, and `customerId`: `2`.

```
InsuranceDB> db.policyAssignments.aggregate([
... {
...   $lookup: {
...     from: "customers",
...     localField: "customerId",
...   }
... }
InsuranceDB> db.policyAssignments.aggregate([
... {
...   $lookup: {
...     from: "customers",
...     localField: "customerId",
...     foreignField: "customerId",
...     as: "customerDetails"
...   }
... }
... ])
... ]
... ]
{
  _id: ObjectId('69574ed259a93b3960ae3c6a'),
  assignmentId: 1,
  customerId: 1,
  policyId: 'P1001',
  agentId: 101,
  startDate: ISODate('2024-01-01T00:00:00.000Z'),
  customerDetails: [
    {
      _id: ObjectId('69574ec459a93b3960ae3c63'),
      customerId: 1,
      firstName: 'Amit',
      lastName: 'Sharma',
      email: 'amit@gmail.com',
      phone: '9876543210',
      city: 'Chennai'
    }
  ]
},
{
  _id: ObjectId('69574ed259a93b3960ae3c6b'),
  assignmentId: 2,
  customerId: 2,
```